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Date:- 27th July, 2026

LO:- LO-04

Experiment-03

Aim:- To deploy multi-service applications using Docker Compose.

Theory:- 1. Mkdir Compose demo

- mkdir stands for make directory.
- It is used to create a new folder in the current working location.
- The folder acts as a workspace to store all the files related to a project.

In the Docker Compose project, the folder Docker-compose-demo contains all the files required to define and run the multi-service application.

2. cd compose demo

changes the current working directory to the folder named Docker-compose-demo.

- cd stands for Change Directory.
- It is a command used in the Command Prompt, PowerShell, Git Bash, and Linux/macOS terminals to move from one folder to another.
- After executing the command, the terminal starts working inside the specified folder.

3. Docker Compose Up

 docker compose is the Docker Compose command-line tool used to manage multi-container applications.

 up is the subcommand that builds (if required), creates, and starts all the containers described in the Compose file.

4. Docker Compose up -d

docker compose up -d is a Docker Compose command used to create and start the services defined in a docker-compose.yml (or compose.yaml) file, running them in the background.

This tells Docker to use Docker Compose, a tool for managing multiple containers as one application stack.

A Compose file describes things like:

- Which images to use
- How containers should be named
- Ports to expose
- Environment variables
- Networks
- Volumes (persistent storage)
- Dependencies between services

5. Docker Compose ps

The command is used to display the status and information of containers managed by a Docker Compose project.

In simple terms:

docker compose ps answers the question: "Which containers belong to my Compose application, and what state are they currently in?"

6. Docker Compose logs

The command is used to view the output logs generated by containers managed by a Docker Compose application.

In simple terms:

docker compose logs shows what is happening inside your Compose services by displaying the messages written by the applications running in those containers.

It is one of the most important commands for debugging and monitoring Docker Compose applications.

7. docker compose stop

The command is used to stop running containers that belong to a Docker Compose application without removing them.

In simple terms:

docker compose stop tells Docker Compose to shut down the running services temporarily while keeping the containers, networks, and configuration available for later restart.

8. Docker Compose Start

The command is used to start existing stopped containers that were previously created by Docker Compose.

In simple terms:

docker compose start turns on containers that already exist. It does not create new containers, rebuild images, or download anything.

9. Docker Compose Restart

The command s used to restart existing containers managed by a Docker Compose application.

In simple terms:

docker compose restart stops running Compose containers and starts them again using the same container configuration.

It is mainly used when you want to refresh services without recreating containers.

10. Docker Compose Down

The command is used to stop and remove the containers, networks, and other resources created by Docker Compose for an application stack.

In simple terms:

docker compose down shuts down a Compose application and cleans up the resources that were created for it.

It is commonly used when you want to completely stop an environment, such as a development setup, test environment, or temporary application stack.

11. Docker Compose Down -v

is used to stop a Docker Compose application, remove its containers and networks, and also delete the volumes associated with that Compose project.

In simple terms:

docker compose down -v completely tears down a Compose environment and removes stored data volumes along with it.

It is a cleanup/destruction command and should be used carefully because it can permanently remove persistent data.

12. Docker Compose Images

is used to display the Docker images used by the containers in a Docker Compose project.

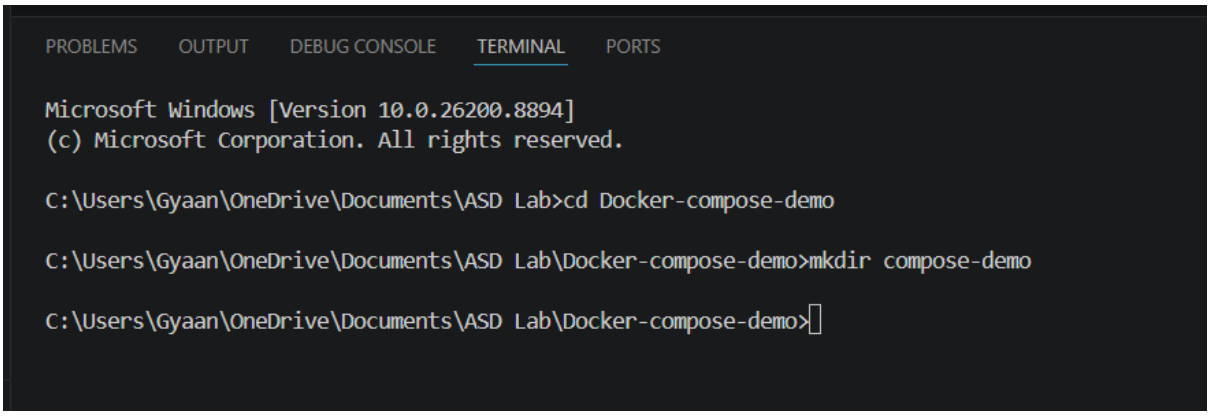
In simple terms:

docker compose images shows you which Docker images are associated with the services defined in your Compose application.

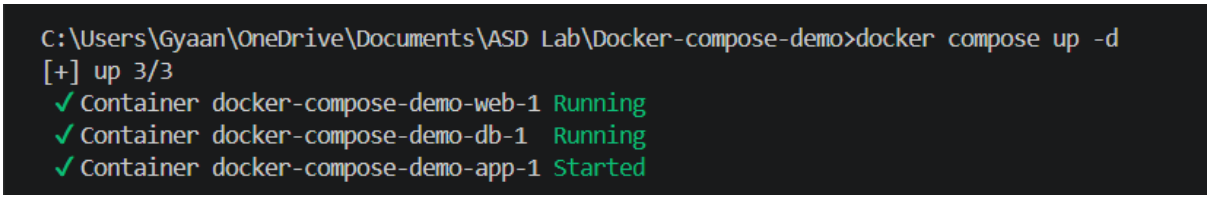
It helps you understand what software packages your Compose containers are running from.

RESULT:-

1. **Mkdir Compose Demo**



2. **Cd Compose Demo**



3. Docker Compose up

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose up
[+] up 29/29
 ✓ Image node:18                        Pulled                                  324.6s
 ✓ Image mysql:8                        Pulled                                  270.3s
 ✓ Network docker-compose-demo_default  Created                                   0.0s
 ✓ Volume docker-compose-demo_db_data   Created                                   0.0s
 ✓ Container docker-compose-demo-web-1  Created                                   0.5s
 ✓ Container docker-compose-demo-app-1  Created                                   0.5s
 ✓ Container docker-compose-demo-db-1   Created                                   0.5s
Attaching to app-1, db-1, web-1
db-1  | 2026-07-27 10:04:08+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.4.10-1.el
9 started.
```

4. Docker Compose Up -d

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose up -d
[+] up 3/3
 ✓ Container docker-compose-demo-web-1  Running
 ✓ Container docker-compose-demo-db-1   Running
 ✓ Container docker-compose-demo-app-1  Started
```

5. Docker Compose ps

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose ps
NAME                       IMAGE          COMMAND                  SERVICE   CREATED         STATUS
       PORTS
docker-compose-demo-db-1   mysql:8        "docker-entrypoint.s…"   db        9 minutes ago   Up 9 mi
nutes   3306/tcp, 33060/tcp
docker-compose-demo-web-1  nginx:latest   "/docker-entrypoint.…"   web       9 minutes ago   Up 9 mi
nutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```

6. Docker Compose logs

```
db-1   | 2026-07-27T10:04:19.521998Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for conn
ections. Version: '8.4.10'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server
 - GPL.
web-1  | 2026/07/27 10:04:08 [notice] 1#1: start worker process 39
web-1  | 2026/07/27 10:04:08 [notice] 1#1: start worker process 40
web-1  | 2026/07/27 10:04:08 [notice] 1#1: start worker process 41
web-1  | 2026/07/27 10:04:08 [notice] 1#1: start worker process 42
web-1  | 2026/07/27 10:04:08 [notice] 1#1: start worker process 43
web-1  | 2026/07/27 10:04:08 [notice] 1#1: start worker process 44
app-1  | Node.js v18.20.8
```

7. Docker Compose Stop

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose stop
[+] stop 3/3
 ✓ Container docker-compose-demo-app-1 Stopped
 ✓ Container docker-compose-demo-db-1  Stopped
 ✓ Container docker-compose-demo-web-1 Stopped
```

8. Docker Compose Start

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose start
[+] start 3/3
 ✓ Container docker-compose-demo-app-1 Started
 ✓ Container docker-compose-demo-db-1  Started
 ✓ Container docker-compose-demo-web-1 Started
```

9. Docker Compose Restart

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose restart
[+] restart 0/3
 - Container docker-compose-demo-app-1 Restarting
 - Container docker-compose-demo-db-1  Restarting
 - Container docker-compose-demo-web-1 Restarting
```

10. Docker Compose Down

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose down
[+] down 4/4
 ✓ Container docker-compose-demo-db-1  Removed
 ✓ Container docker-compose-demo-web-1 Removed
 ✓ Container docker-compose-demo-app-1 Removed
 ✓ Network docker-compose-demo_default Removed
```

11. Docker Compose Images

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>docker compose images
CONTAINER           REPOSITORY          TAG                 PLATFORM            IMAGE ID
IZE                 CREATED
```

12. Docker Compose Config

Document.yml

```
services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"

  app:
    image: node:18
    working_dir: /app
    volumes:
      - ./app:/app
    command: node server.js

  db:
    image: mysql:8
    environment:
      MYSQL_ROOT_PASSWORD: password
      MYSQL_DATABASE: mydb
    volumes:
      - db_data:/var/lib/mysql

volumes:
  db_data:
```

13. Dir

```
C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo>dir
 Volume in drive C is OS
 Volume Serial Number is 14BF-6742

 Directory of C:\Users\Gyaan\OneDrive\Documents\ASD Lab\Docker-compose-demo

27-07-2026  15:35    <DIR>          .
27-07-2026  15:17    <DIR>          ..
27-07-2026  15:18    <DIR>          app
27-07-2026  15:35    <DIR>          compose-demo
27-07-2026  15:28               371 compose.yaml
```

CONCLUSION:-

Through the study of Docker Compose commands, we have learned how Docker simplifies the process of creating, managing, monitoring, and maintaining containerized applications. Docker Compose provides a structured way to define multiple services, configure their relationships, and control the complete application lifecycle using simple commands.

Docker Compose commands are not just individual commands; they represent different stages of managing a containerized environment. Each command performs a specific responsibility, and together they form a complete workflow for application deployment and maintenance.